Pengyu Zhang

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University of Amsterdam, The Netherlands Fourth-year Ph.D. Candidate

Education

2022.10 – 2026.10 **Ph.D. in Computer Science**, Intelligent Data Engineering Lab (INDE Lab), University of Amsterdam (UvA), Amsterdam, The Netherlands.
Supervisors: Prof. Dr. Paul Groth, Dr. Klim Zaporojets.

2019.09 – 2022.07 **M.Eng. in Control Engineering**, Beijing University of Technology (BJUT), Beijing, China.
Supervisor: Prof. Dr. Yong Zhang.

Research Interests

My research focuses on building robust knowledge representations in evolving knowledge graphs by fusing multiple information sources, including graph structure, text, images, and temporal signals. I develop methods that address temporal degradation and multi-modal integration across tasks such as knowledge graph completion and graph-based recommendation.
Keywords: Knowledge Graphs, Entity Linking, Multi-Modal Knowledge Graphs, Graph Neural Networks, Contrastive Learning, Multi-Modal Fusion

Publications

2026 **Beyond Images: Are a Thousand Words Better Than a Single Picture? A Framework for Multi-Modal Knowledge Graph Dataset Enrichment**
Pengyu Zhang, *Klim Zaporojets, Congfeng Cao, Jie Liu, and Paul Groth.*
In *The Extended Semantic Web Conference (ESWC 2026).*
Paper Code

2025 **A Survey of Large Language Models for Data Challenges in Graphs**
Mengran Li, ***Pengyu Zhang****, Wenbin Xing, Yijia Zheng, Klim Zaporojets, Junzhou Chen, Ronghui Zhang, Yong Zhang, Siyuan Gong, Jia Hu, Xiaolei Ma, Zhiyuan Liu, Paul Groth, and Marcel Worring.*
In *Expert Systems with Applications.*
Paper Code

2024 **Understanding the Impact of Entity Linking on the Topology of Entity Co-occurrence Networks for Social Media Analysis**
James Nevin, ***Pengyu Zhang****, Dimitar Dimitrov, Michael Lees, Paul Groth, and Stefan Dietze.*
In *International Conference on Knowledge Engineering and Knowledge Management (EKAW 2024).*
Paper Code

CYCLE: Cross-Year Contrastive Learning in Entity-Linking
Pengyu Zhang*, Congfeng Cao, Klim Zaporojets, and Paul Groth.*
In *ACM International Conference on Information and Knowledge Management (CIKM 2024).*
Paper Code

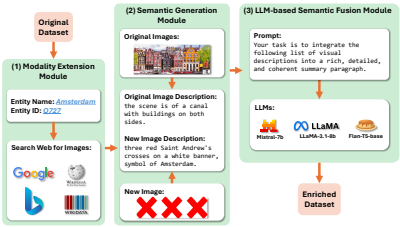
TIGER: Temporally Improved Graph Entity Linker
Pengyu Zhang*, Congfeng Cao, and Paul Groth.*
In *European Conference on Artificial Intelligence (ECAI 2024).*
Paper Code

Publications (continued)

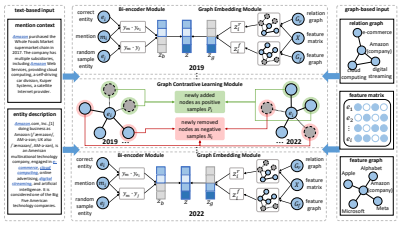
Relationship Updating Network with Contrastive Learning
Pengyu Zhang, *Yong Zhang, Xinglin Piao, Yongliang Sun, and Baocai Yin.*
In *Physica A: Statistical Mechanics and its Applications*.
Paper Code

2023 **MVMA-GCN: Multi-view Multi-layer Attention Graph Convolutional Networks**
Pengyu Zhang, *Yong Zhang, Jingcheng Wang, and Baocai Yin.*
In *Engineering Applications of Artificial Intelligence*.
Paper Code

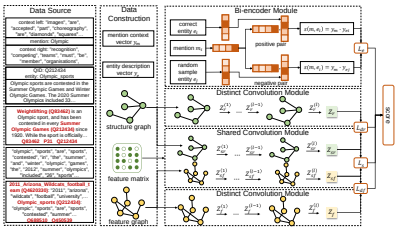
Project Experiences



Beyond Images (ESWC 2026).
Problem: MMKG construction suffers from limited image coverage and ambiguous visuals (e.g., logos, abstract art), reducing downstream robustness.
Method: We build an automated enrichment pipeline that retrieves additional web images, converts images to textual descriptions via vision-language models, and fuses descriptions with an LLM into entity-aligned summaries.
Outcome: The enriched data boosts performance by up to +7% Hits@1 overall and +333% Hits@1 on logo/symbol subsets.



CYCLE: Cross-Year Contrastive Learning in Entity-Linking (CIKM 2024).
Problem: Entity-linking models degrade as knowledge graphs evolve over time, especially with structural relationship shifts across yearly snapshots.
Method: We propose graph contrastive learning across temporal snapshots, using newly added relationships as positives and removed relationships as negatives, and construct the GCL-TempEL dataset with Wikidata5M relations.
Outcome: CYCLE improves over prior state-of-the-art by 13.9% at a 1-year gap and 17.8% at a 3-year gap, with strong gains for low-degree entities.



TIGER: Temporally Improved Graph Entity Linker (ECAI 2024).
Problem: Temporal degradation hurts entity-linking performance as graph structure and textual evidence drift over time.
Method: TIGER combines distinct/shared graph convolution modules for structure and feature graphs with text bi-encoder representations, and introduces the Graph-TempEL benchmark.
Outcome: TIGER achieves +16.24% over SpEL at a 1-year gap and +20.93% at a 3-year gap, with notable gains on high-degree entities.